

ST6003CEM Web API Development

Project Proposal on nepalink



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Table of contents

Table of figure

Introduction

Imagine being thousands of miles apart from your aging parents and always fearing that they took their medication or had a good meal. This is what happens to millions of families in the modern world that globalization and urban migration make loved ones distant in their cities and countries including Nepal. Caring effectively about the elderly members of the family becomes a problem, which is usually the cause of stress and irrespectiveness.

To this emerging crisis, Nepalink offers a new Caregiver Hiring and Task Management System - an application that aims at bridging the gap between family members and the reliable caregivers to provide an easy way of doing the care giving. Nepalink provides a safe and convenient platform, in which members will have the opportunity to hire or fire caregivers, delegate duties, follow-ups, and engage in smooth communication.

In this proposal, I am planning to create the backend system, admins dashboard, and end-user interface of Nepalink, using the concept of modern technologies as the MERN stack (MongoDB, Express.js, React, and Node.js) to develop a scalable, reliable, and secure system. The mission of Nepalink is to eliminate the distance/care gap and provide a sense of peace of mind to the family members and good services to the caregivers.

Justification

Families are somewhat separated due to all the business, school, and transportation which makes it difficult to watch after our elderly parents. How people typically care about them would be simply casual, e.g., they lack strict regulations, they do not engage in any form of accountability to one another and is terribly slow. The family members are unable to keep track of what is done by whom, whether the job is done and keep in touch with each other, particularly when one is abroad.

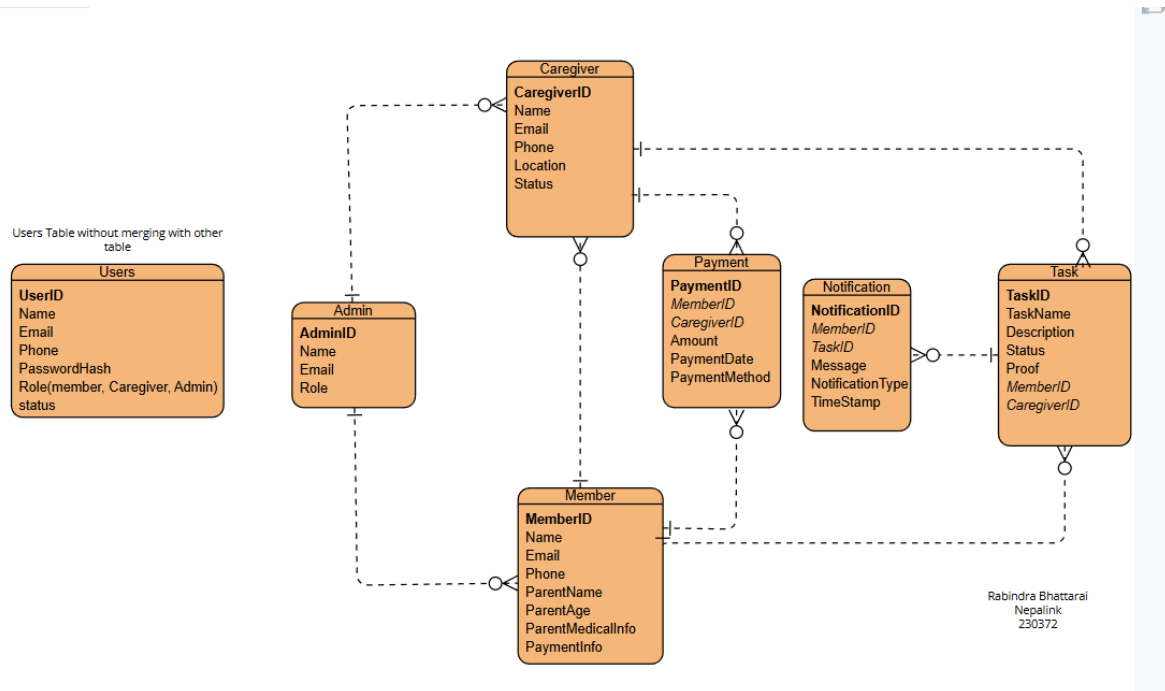
NepaLink assists in organizing all of that by providing a digital platform that makes recruiting and maintaining caregivers extremely easy. It makes it real through live tracking, secure chat and the in built payment, as well as their GPS sharing keeps everybody safe. Going digital with the entire process will allow families to rest easy and caregivers the opportunity to know exactly what will be required of them.

Objectives

- Build a secure and strong backend using Node.js and Express.js.
- Create a simple and responsive web platform for members, caregivers, and admins.
- Allow members to hire or remove caregivers, add tasks, and check progress easily.
- Give caregivers the option to see tasks, mark them as done, and upload photo or video proof.
- Add safe login and account security using JWT and OTP.
- Include online payment so members can pay caregivers without hassle.
- Show real-time updates for tasks, caregiver location, and payments.
- Add useful features like chat, location tracking, and task history for better communication and trust.

ERD

The ER diagram of Nepalink presents the principal entities: Member, Caregiver, Admin, Task, Payment and Notification. The members are able to employ Caregivers, delegate, caregivers report status of the task and location, and the admins go through all the accounts and system information.



After Project Completion project Mission

When we are finished, Nepalink hopes to become the highest-rated caregiver services site in Nepal and it will expand to the global scene in the future. This is aimed at assisting the caregivers to increase their on-line presence and provide families with a reliable, accessible, and understandable means of managing care on behalf of their loved ones.

Scope of work (Technologies Used)

The project will cover the development of a complete caregiver management platform, including:

- Backend with secure and scalable operations with the help of Node.js and Express.js.
- MongoDB database that will store user, task and payment information.
- React.js For frontend development of responsive member, caregiver and assistant administrator dashboards.
- The usage of Google Maps to track the whereabouts of caregivers in real time.
- Easy money transferred via payment gateways (eSewa, Khalti, Stripe).
- Task update and alert notifications system.

Features of the project

The Nepalink platform will include:

- A secure backend written in Node.js and Express.js to ensure that all is smooth.
- Member, caregiver, and administrator dashboards are appealing to the phone and computer, both designed using React.js.
- An activity management tool that allows assigning gigs, tracking your progress, and staying on top of everything.
- Google Maps live location tracking to be able to know where caregivers are at any given time.
- Quick and easy money movements through the use of eSewa, Khalti and Stripe.
- A chat tool that will allow members to speak to caregivers in real-time.
- Task update and payments notifications and alerts to ensure that nobody is left out.

Prototypes

The prototypes of the Nepalink were scaled down to Figma and everything is clean, user-friendly. It received screens on the Member Dashboard, Caregiver Dashboard and Admin Panel and featured items such as task management, messaging, payment integration and location monitoring.

Deliverables

- Complete back-end implementer, API documentation and all, in place.
- Member, caregiver, and administration interactive dashboards, such as completely user-friendly.
- Safe access and payment connection- makes it secure.
- Database schema together with the initial set up--no headaches.
- The locked down source resides in an opaque repository.
- Install and run the entire platform- so that it is functioning.

Estimated timeline

To develop Nepalink, I am going to design, write code, test and finally deploy it. It will require approximately 2 to 3 months which will include the backend, frontend dashboards, integrating payment and location services and lastly going live.

Conclusion

Nepalink is designed to address the growing need for reliable caregiving solutions for families living away from their loved ones. By integrating features such as task management, location tracking, secure communication, and payment processing, the platform ensures transparency and convenience for both members and caregivers.

Built on modern technologies like the MERN stack, Nepalink offers a scalable and secure solution that bridges the gap between distance and care. Upon completion, this system will not only simplify caregiving but also establish a trusted digital platform that empowers caregivers and provides peace of mind to families.

Reference